

Node.js

Digital Notebooks

File System & JSON APIs

Learn to save your code's memory forever!



The Memory Problem

Variables are like things we hold in our hands.
When we stop working, they vanish!
Your computer's RAM is like a brain that forgets
everything when it goes to sleep.

How can we save our notes forever?
What if we want our data to stay safe
even after we close our program?



The Digital Almirah

The fs Module



File System = Your Almirah

Just like your cupboard at home stores clothes safely, the fs module stores your data!



Data Stays Safe

Even when your computer is turned off, your saved files remain in the almirah!



Permanent Storage

Unlike variables that disappear, files in fs module stay forever until you delete them.



Writing Files with `fs.writeFileSync()`

```
1 // Node.js
2 // Example: Writing a file synchronously
3 // Description: This script writes a file named 'notes.txt' with the content 'My first note!'.
4 // It uses the fs module and the writeFileSync method.
5
6 // Import the fs module
7 const fs = require('fs');
8
9 // Write the file synchronously
10 fs.writeFileSync('notes.txt', 'My first note!');
11
12 // Log a message to the console
13 console.log('File written successfully!');
```

`const fs = require('fs');`
`fs.writeFileSync('notes.txt', 'My first note!');`
Save your notes forever with `writeFileSync`!



```
1 success.test {
2   {fn}.file(JestinScore.txt, notes);
3   eOres caa" -tut{lnex" - "fs.readFileSync" - utf8);
4   hof a tancmler"y?
5   tarf all JileVern= uut.th0 {
6     ns-tignsfilter"= tq t"}
7   Lour all.tme{"ino cetter. ()};
8   óód snroërseSC)
13 }
9   ny lécasfiletcl{& etscr.yt(0 )}
1   fef lsl, console log }
11 }
13 bead a otyotier== : frole"
4   ny mead dstl { "pup cettira(0MatclF));
5   Cond a cayli { } "aup catiic atlt= charety,);
6   lsa-y.aylls{ fff! }
7   }
12 }
```

Reading Files with fs.readFileSync()

Load your saved notes anytime! Use `readFileSync` to retrieve data from files and display it in your program.

RECIPE CARDS

INGREDIENTS:



STEPS 2:



Meet JSON: The Recipe Format for Data

Think of JSON like a recipe card for Maggi or Biryani — it has clear labels like "Ingredients" and "Steps" that everyone can read and understand the same way!

Why JSON is useful:

- Universal format — computers everywhere understand it
- Organized data with clear labels
- Easy to read for both humans and machines
- Perfect for saving structured information!



Packing vs Unpacking

JSON.stringify & JSON.parse

**JSON.stringify = Packing a suitcase
(converting object to string for storage)**

**JSON.parse = Unpacking the suitcase
(string back to usable object)**

Example: { name: "Asha", grade: 6 }

Creating a Student Profile with JSON



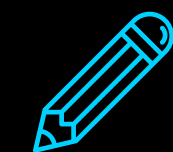
Create Student Object

`const student = { name: "Rahul", rollNo: 15, favSubject: "Math" };`



Convert to JSON String

`const jsonStr = JSON.stringify(student);`
packs your data into a neat format!



Save to File

`fs.writeFileSync('student.json', jsonStr);`
stores it in your Digital Almirah!

```
1  nuth.jeducacton.ls instudent-Parrile.713:OSs.-n Fd sections ->
2  print "const(rvpist:)
4
5      name: {student};
6      favSubject: Math;
6      rollNo: 15;
9      favSubject: Math;
11     favSubject: =>;
19     JSON.stringify(student);
10
13
14     fs.writeFileSync,
15
17
```


Activity Time:

Build Your Digital Tiffin Diary!

Task: Write your lunch menu for the week and save it in tiffin.json!

Sample JSON:

```
{  
  "Monday": "Idli with Sambar",  
  "Tuesday": "Aloo Paratha",  
  "Wednesday": "Flattened rice dish",  
  "Thursday": "Spicy chickpeas"  
}
```

Let's save your tasty tiffin notes!



JSON

Scavenger Hunt!

Find the Mistakes

Can you spot the errors in this broken JSON?
{ name: Rahul, rollNo: 15 favSubject: "Math" }
Hint: Look for missing quotes and commas!
Fix the JSON to make it work properly.



Recap:

.txt vs .json Files

.txt Files

Simple notes, no structure

Just plain text

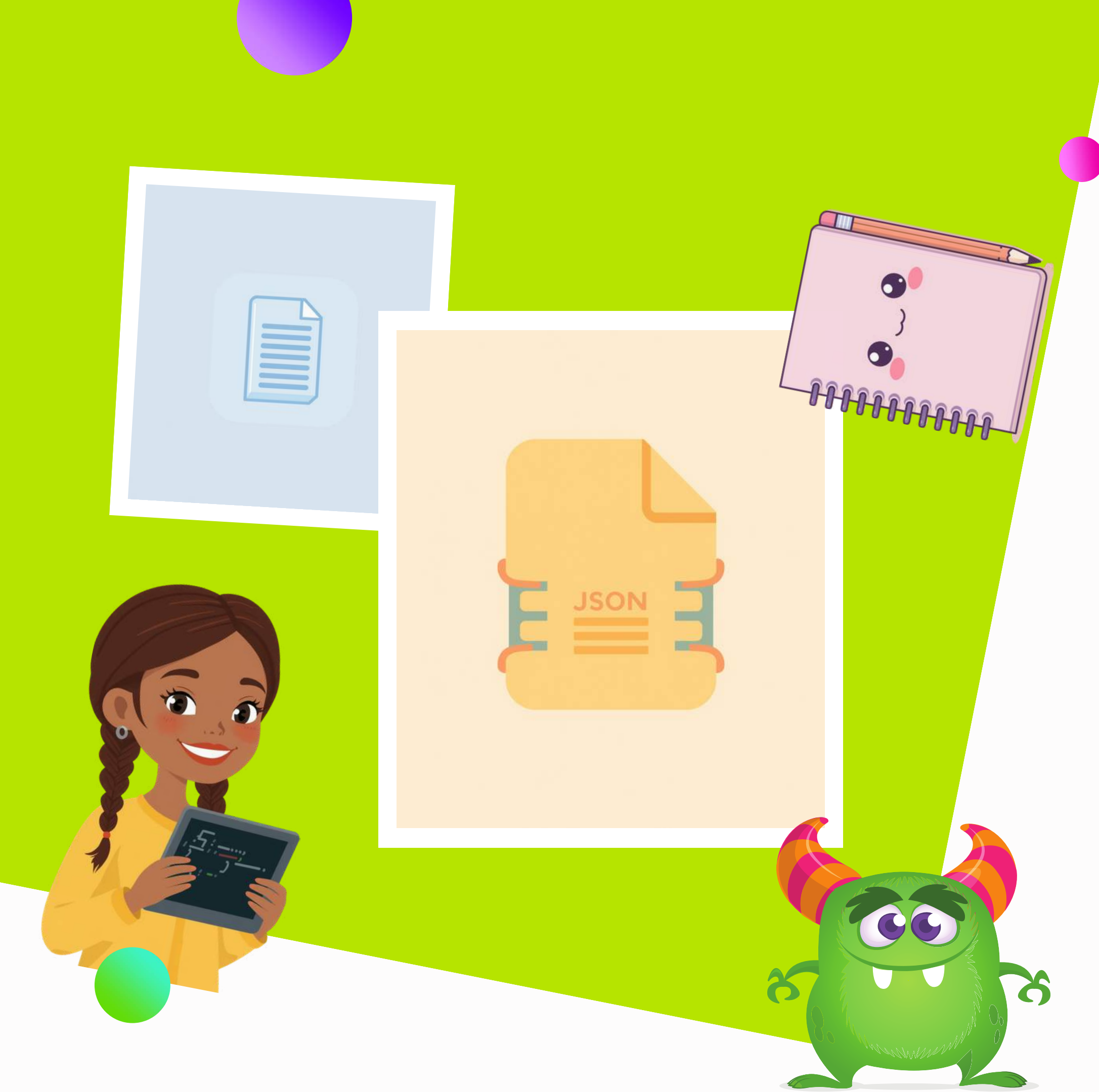
Easy to read but no labels

.json Files

Data with labels and structure

Easy for computers to understand

Perfect for saving information with meaning!



Key Takeaways From Today's Session



Digital Almirah

File System lets us save data permanently — like a cupboard that keeps your notes safe!



Recipe Format

JSON is a universal format to store structured data — everyone reads it the same way!



Pack & Unpack

We can pack (stringify) and unpack (parse) data easily — your digital notebook stays safe and ready!

Keep Exploring!

Your Code's New Superpower

"We are giving our code a long-term memory!"

What will you create next with your Digital Notebook? Your programs can now remember everything — notes, recipes, student profiles, and more!





Thank You!

Any Questions?

Thank you for joining today's session on Digital Notebooks!
We hope you enjoyed learning about File System & JSON APIs.



Next Session: Coming Soon!

Keep coding and building your Digital Notebooks!

